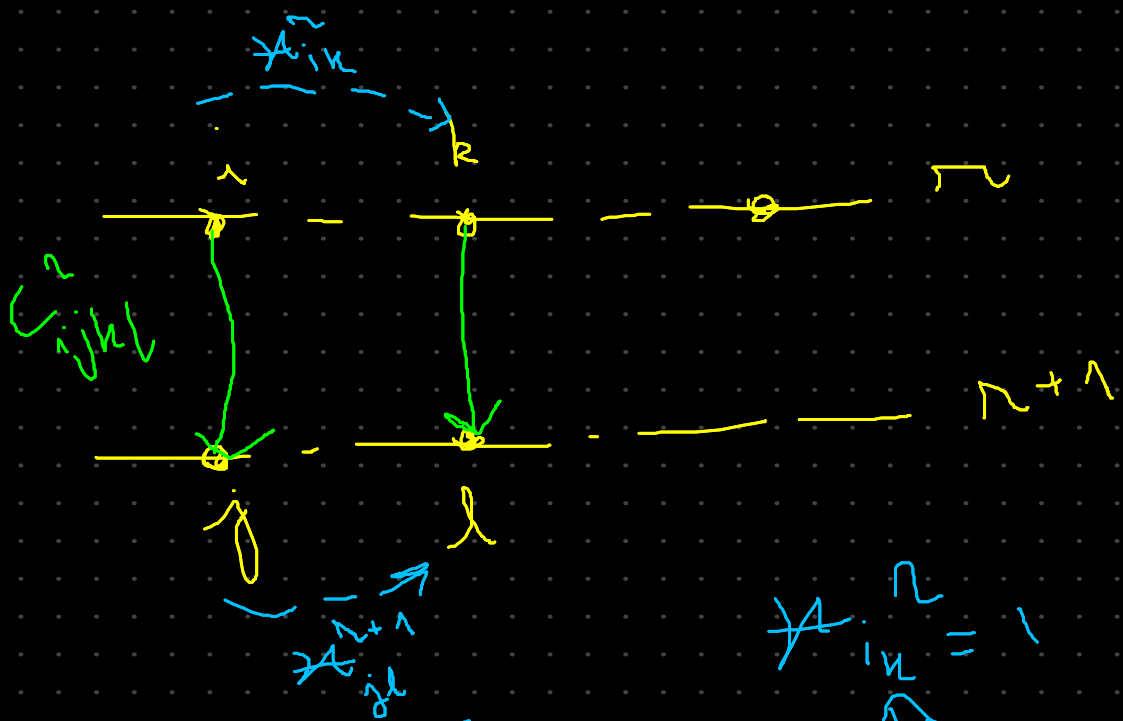




$$C_{ijkl}^n = \left[ \begin{array}{c|c} x_{ik}^n & x_{ej}^{n+1} \\ \hline x_{hi}^n & x_{jl}^{n+1} \end{array} \right]$$

$$\begin{aligned} x_{ik}^n &= 1 \\ x_{hi}^n &= 0 \end{aligned} \quad (2)$$

$$0 \leq x_{il}^n \leq C_{ijkl}^n$$

since  $n \rightarrow 1$

$$0 \leq x_{il}^n \leq C_{ijkl}^n$$

$$\Rightarrow |x_{il}^n - x_{jl}^{n+1}| \leq C_{ijkl}^n \quad (6)$$

$$1 - C_{ijk}^n \leq x_{jk}^{n+1} + x_{ik}^n \leq 1 + C_{ijk}^n$$

$$0 \leq x_{jk}^{n+1} - x_{ik}^n \leq C_{ijk}^n$$

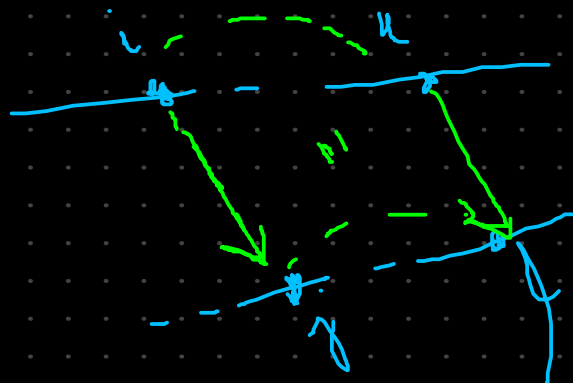
$$0 \leq x_{jk}^{n+1} - 1 + x_{ik}^n \leq C_{ijk}^n \quad (*)$$

$$1 - C_{ijk}^n \leq x_{jk}^{n+1} + x_{ik}^n \leq 1 + C_{ijk}^n$$

$$x_j^{n+1} = x_{ik}^n, (i,j), (k,l) \in E_n, j < i$$

Lemma 2:  $C_{ijk}^n = 0$  is necessary

$$x_{jk}^{n+1} = x_{ik}^n, (i,j), (k,l) \in E_n$$



$$c_{ijk} \in \{0, 1\}$$

$$c_{ijk} \{0 \leq A_{ik} \leq c_{ijk}\}$$

$$|n - A_{ik}| \leq c_{ijk}$$



$$x_{ab} = x_{ci} + x_{ej} = x_{ab}$$

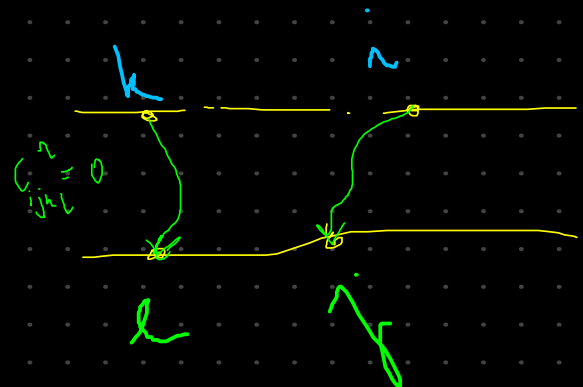
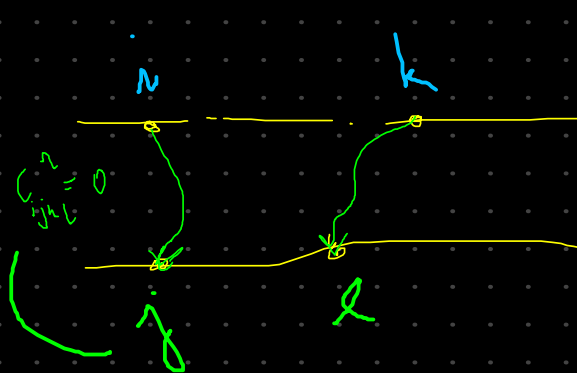
$$x_{ab} \neq$$

$$x_{ab}$$

$$x = j \rightarrow x$$

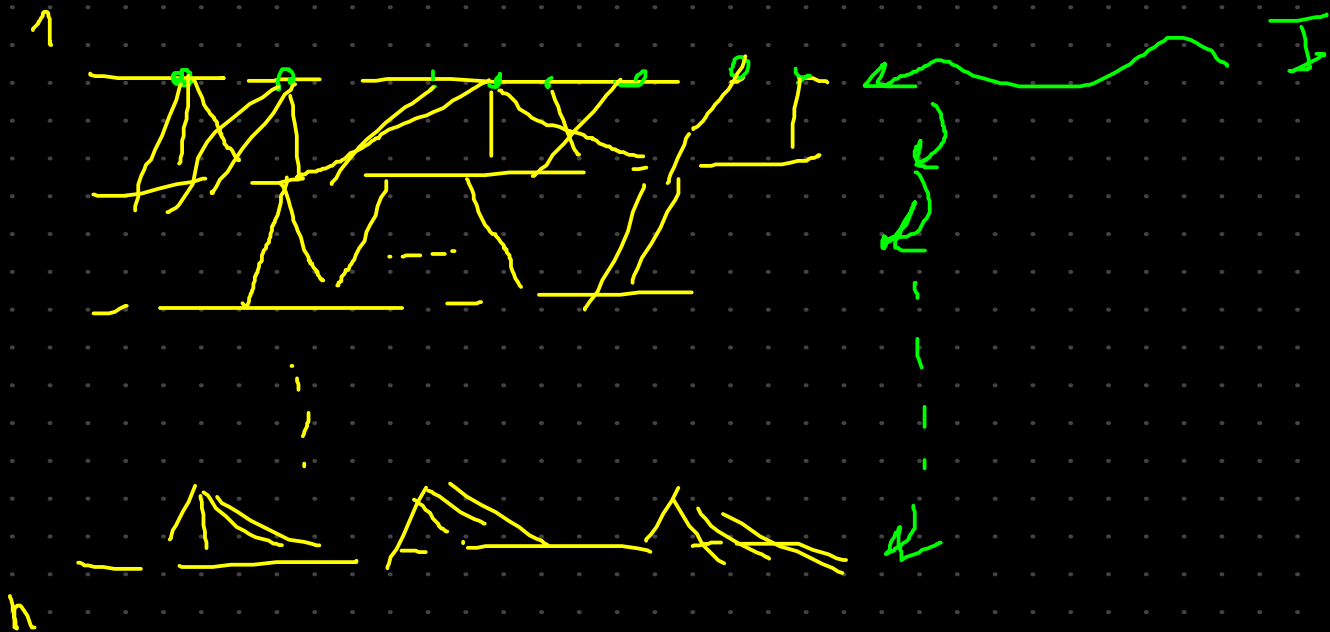
$$x \neq y \neq j \neq c \neq l \neq y$$

$$k \neq e \neq f \neq k \Rightarrow \underline{k \neq k}$$



~~Inputs~~ it thinks that's the output

If I understand correctly, building a level planar graph with this heuristic would need to start from the top level:



set of constraints

$$x = y$$

$$x \neq y$$

$$\mathcal{C} = (x_1 = x_2, x_2 \neq x_3, \dots, x_i = x_{i+1})$$

if we can't derive an <sup>inconsistent</sup> closed chain from  $\mathcal{C}$

$$\left( \begin{array}{l} x = y \\ \equiv \\ \overline{x} \end{array} \right) \left( \begin{array}{l} x \neq y + 2 \\ \equiv \\ \overline{x} \end{array} \right)$$

$\mathcal{C}$  can be satisfied by  
 $\{0, 1\}$ -assignment

we can assign  $\{0, 1\}$  to each  
variable  $x_i$  in  $\mathcal{C}$ .

— level graph planarity  
and layout

$$\sqrt{\pi_{ij}^2} + \sqrt{\pi_{kl}^2}$$

$$\equiv \pi_{ij}^2 \neq \pi_{kl}^2$$

